

Printing solar cells like newspaper

Scientists at Chemnitz University of Technology (Germany) have developed roll-to-roll printed solar cells that can be

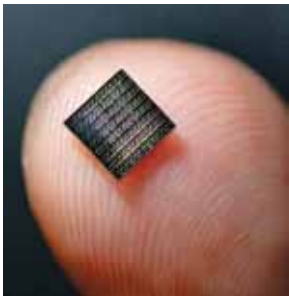


Researchers at Chemnitz University of Technology test-print a conductive PEDOT:PSS layer for organic solar cells using a roll-to-roll press (Credit: <https://interestingengineering.com>)

produced on paper or plastic, similar to newspapers. Using a polymer blend called PM6:Y12, the flexible cells now achieve 9% efficiency and an 88% production yield, up from 1.7% in 2011. The low-temperature printing process reduces energy consumption and costs, paving the way for scalable, lightweight solar panels that can be integrated into everyday surfaces and portable energy applications.

Brain-like artificial neurons for AI chips

Researchers at the University of Southern California have developed artificial neurons that mimic the ion-driven



The chip of an array of integrated artificial neurons, fabricated in USC cleanroom (Credit: <https://viterbischool.usc.edu>)

signalling of real brain cells, using a 'diffusive memristor' instead of conventional transistors. Each neuron functions with the footprint of a single transistor, drastically reducing chip size and energy use. The design, which processes information through ion movement like the human brain, could lead to ultra-efficient neuromorphic chips and future brain-like AI hardware.

C-shaped diode for terahertz waves

Researchers at the University of Electronic Science and Technology (China) have designed an asymmetric C-shaped Schottky diode chain that boosts terahertz frequency multiplication efficiency. By redistributing local electromagnetic fields, the structure ensures uniform energy flow across the diodes, thereby overcoming the power-handling limits of traditional designs. Tests showed a 38% frequency-dou-

bling efficiency and over 300mW output at 170GHz, marking a step toward compact, high-power terahertz sources for imaging and communication systems.

India demonstrates quantum hardware and cell therapy achievements

At the Emerging Science, Technology and Innovation Conclave (ESTIC) 2025, New Delhi, India unveiled three key breakthroughs: the Quantum Secure Information Processor



India unveils new research in quantum hardware and cell therapy at ESTIC 2025, addressing challenges in secure computing, data encryption, and targeted cancer treatment (Credit: www.pib.gov.in)

(QSIP), a 25-qubit Quantum Processing Unit (QPU), and NexCAR19, the nation's first indigenous CAR-T cell therapy. Together, they tackle challenges in quantum-era cybersecurity, high-performance computing, and cancer treatment. The initiatives mark India's growing capability in developing quantum-safe chips, scalable superconducting processors, and affordable, humanised immunotherapies.

Ultra-low-power MCUs for wearables

Researchers at ETH Zurich (Zürich, Switzerland) have developed NanoHydra, a system that enables ultra-low-power microcontrollers (MCUs) in wearable devices to efficiently run complex machine learning tasks. Designed for time series classification, such as ECG or motion analysis, NanoHydra replaces heavy floating-point math with lightweight binary operations, reducing energy use by 18 times while maintaining over 94% accuracy. Tested on GreenWaves' GAP9 chip, it processed ECG data in 0.33ms, enabling wearables to run for years on a single battery.

Vertically stacked transistors for flexible and high-density electronics

Researchers at University of Science and Technology (KAUST), Thuwal, Kingdom of Saudi Arabia have developed a new method to vertically stack transistors for flexible and high-density electronics without the usual heat and defect issues. The process alternates between N-type oxide and P-type organic transistors across six layers, totaling 41, using smooth, insulating parylene C films between them. Tested across 600 devices, the design delivers balanced performance and low-power operation, enabling compact, flexible circuits for wearables and IoT systems while remaining compatible with large-area fabrication.